

# Ali Alemami

Systems & Backend Developer

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**Computer Science student specializing in backend & low-level systems engineering**, with hands-on experience in **C/C++** (POSIX APIs, multithreading, memory management) and **C#/NET** (3-tier architecture, SQL Server).

## TECHNICAL SKILLS

|                                    |                                                                                                                              |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages:</b>      | C (POSIX/C99), C++ (98/11/17), C# (.NET), SQL (T-SQL), Bash                                                                  |
| <b>Systems &amp; OS Concepts:</b>  | POSIX APIs, Multithreading (pthreads), Concurrency & Mutexes, Sockets / epoll, Process Control & IPC, Memory (Valgrind/ASan) |
| <b>CS Fundamentals:</b>            | Data Structures & Algorithms, OOP & Design Patterns, 3-Tier Architecture, Computer Networks                                  |
| <b>Tools &amp; Infrastructure:</b> | SQL Server, ADO.NET, Docker, Docker Compose, NGINX, Git, Linux, GDB, Makefiles                                               |
| <b>Languages (Spoken):</b>         | Arabic (Native), English (Fluent / Professional)                                                                             |

## EXPERIENCE

**42 Amman** — Full-time intensive program, peer-reviewed project development 2025 – Present

*Structured, deadline-driven software engineering program — 22/24 Common Core projects complete*

- **Multithreading & Concurrency** (C — [philo](#)): Engineered a race-free simulation of Dijkstra's Dining Philosophers using pthreads and mutex locks; built async monitoring routines with microsecond-precision timers preventing thread starvation and deadlocks.
- **HTTP Web Server** (C++98 — [webserv](#)): Building a non-blocking HTTP/1.1 server using epoll I/O multiplexing, with stateful request parsing, NGINX-style config parsing, and CGI execution.
- **DevOps & Container Infrastructure** ([inception](#)): Architected a multi-container infrastructure with Docker Compose on Alpine Linux; isolated containers for NGINX (TLS/SSL), WordPress, and MariaDB with bind mounts and internal bridge networking.
- **Unix Shell & AST Engine** (C — [minishell](#)): Implemented a POSIX-compliant shell with an AST parser enforcing operator precedence, nested subshell forks, and quote-aware tokenization; zero leaks via Valgrind.
- **Unix Process Pipelines & IPC** (C — [pipeX](#)): Engineered a Unix pipeline engine replicating shell redirection (`< file1 cmd1 | cmd2 > file2`) using fork, pipe, dup2, and execve; implemented POSIX process synchronization, multi-fd stream redirection, PATH resolution, and error handling.
- **Algorithm Optimization & Dual-Stack Sorting** (C — [push\\_swap](#)): Engineered an optimized dual-stack sorting engine with cache-friendly array stacks and coordinate compression bounding bit passes to  $\log_2(N)$ ; designed a lookahead binary radix sort with predictive next-bit retention in Stack B eliminating redundant transfers, paired with heuristic solvers for  $N \leq 5$ .
- **C++ Modules & Ford-Johnson Algorithm** (C++ — [cpps](#)): Completed 42 C++ Modules 00–09 (OOP, polymorphism, templates, STL); implemented Ford-Johnson Merge-Insert Sort optimizing comparisons via Jacobsthal sequence pairing.
- **Foundations & Algorithms**: [mini\\_rt](#) (3D raytracer with Phong illumination), [fract-ol](#) (fractal rendering), [minitalk](#) (UNIX signal IPC), [get\\_next\\_line](#), [ft\\_printf](#), [libft](#).
- **Peer Review & Collaboration**: Conducted 50+ technical evaluations and collaborated in pair-programming teams ([minishell](#), [mini\\_rt](#)).

## SOFTWARE DEVELOPMENT & DATABASE PROJECTS

*Programming Advices roadmap (Dr. Mohammed Abu-Hadhoud) — 17/24 courses completed in C#/NET, SQL Server, and C++*

- **Bank & ATM CLI System** (C++ — [Bank-and-ATM-System-CPP](#)): Built a terminal-based banking and ATM system in C++ with bitmask user permissions, transaction audit logging, and file-based persistence.
- **Contacts Management App** (C#, ADO.NET, SQL Server — [Contacts-App-3Tier](#)): Developed a 3-tier desktop application separating UI, business logic, and data access layers; used parameterized T-SQL queries to prevent SQL injection.
- **Relational Database Schemas & SQL** (SQL Server — [Programming-Advices-Core](#)): Designed 5 relational database schemas (Medical Clinic Booking, E-Commerce Store, HR/Payroll, Vehicle Registry, Employee Analytics) with normalized keys and multi-table JOIN queries.
- **Algorithms & Problem Solving** (C++ — 225+ Problems): Implemented solutions across 5 difficulty levels covering data structures, matrix operations, string parsing, and combinatorial algorithms.
- **Windows Forms Projects** (C#, .NET — [CSharp-WinForms-Collection](#)): Built 10+ desktop utility and game apps in C#, focusing on UI event handling and object-oriented design.

## EDUCATION

**The Hashemite University** — BSc in Computer Science

2022 – 2026 (Expected)